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ELECTRO-OPTICAL DEVICE AND ELECTRONIC APPARATUS

Abstract

An electro-optical device includes a pixel electrode, a first wiring line including a first layer, a second layer, and a third layer, a transistor at least partially overlapping the first wiring line in plan view, and a data line provided in a layer between the first wiring line and the transistor, at least a portion of the data line overlapping the first wiring line in plan view, and electrically coupled to the pixel electrode via the transistor, in which the first layer has light reflectance and is disposed closer to the transistor than the second layer and the third layer, the second layer is provided between the first layer and the third layer and has light reflectance lower than the first layer, and the third layer has light reflectance higher than the second layer.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-026439, filed Feb. 26, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.
BACKGROUND
1. Technical Field
[0002] The present disclosure relates to an electro-optical device and an electronic apparatus.
2. Related Art
[0003] Typically, as one type of electro-optical device, an active-drive type liquid crystal device is known in which each pixel is provided with a transistor that controls switching of a pixel electrode. Such a liquid crystal device is provided with a contact hole used to couple various types of wiring lines or electrodes or the like. For example, JP-A-2005-242296 discloses an electro-optical device. This electro-optical device includes a contact hole used to couple a shield layer and a relay layer for the shield layer, and also includes a contact hole used to couple a third relay electrode and a second relay electrode.
[0004] In the electro-optical device disclosed in JP-A-2005-242296, when a stacked structure is adopted in the entire region of the contact hole, the shield layer, and the relay electrode, for example, a material having high light absorption efficiency such as titanium nitride may be used as an electrolytic corrosion prevention layer. However, when titanium nitride absorbs light, there is a problem that heat is likely to be generated inside a substrate.
SUMMARY
[0005] In order to solve the above-mentioned problems, an electro-optical device according to an aspect of the present disclosure includes a pixel electrode, a first wiring line including a first layer, a second layer, and a third layer, a transistor at least partially overlapping the first wiring line in plan view, and a data line provided in a layer between the first wiring line and the transistor, at least a portion of the data line overlapping the first wiring line in plan view, and electrically coupled to the pixel electrode via the transistor, in which the first layer has light reflectance and is disposed closer to the transistor than the second layer and the third layer, the second layer is provided between the first layer and the third layer and has light reflectance lower than the first layer, and the third layer has light reflectance higher than the second layer.
[0006] An electronic apparatus according to another aspect of the present disclosure includes the electro-optical device according to the above-mentioned aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is a schematic plan view illustrating a configuration of a liquid crystal device according to a first embodiment.
[0008] FIG. **2** is a schematic cross-sectional view illustrating a structure of the liquid crystal device in a line segment A-A′ in FIG. **1**.
[0009] FIG. **3** is an equivalent circuit diagram illustrating an electrical configuration of the liquid crystal device.
[0010] FIG. **4** is a schematic plan view illustrating the arrangement of pixels.
[0011] FIG. **5** is a cross-sectional view illustrating a structure of an element substrate in a line

segment β**1**-β**2** in FIG. **4**.

[0012] FIG. **6** is a cross-sectional view illustrating a structure of an element substrate in a line segment α**1**-α**2** in FIG. **4**.

[0013] FIG. **7** is a cross-sectional view illustrating a structure of an element substrate in a line segment γ**1**-γ**2** in FIG. **4**.

[0014] FIG. **8** is a plan view illustrating a configuration of common wiring lines and third relay electrodes.

[0015] FIG. **9** is a plan view illustrating a mode of a third relay electrode according to a second embodiment.

[0016] FIG. **10** is a cross-sectional view illustrating a structure of an element substrate in a line segment β**1**-β**2** in FIG. **4**.

[0017] FIG. **11** is a cross-sectional view illustrating a structure of an element substrate in a line segment α**1**-α**2** in FIG. **4**.

[0018] FIG. **12** is a schematic view illustrating a configuration of a projection-type display device according to a third embodiment.

DESCRIPTION OF EMBODIMENTS

[0019] Below, embodiments according to the present disclosure will be described with reference to the drawings. The present disclosure is not limited to the following embodiments. Various types of modification examples that are implemented are also included in the present disclosure within a range in which the main points of the present disclosure do not change.

[0020] In the following individual drawings, X, Y, and Z axes that represent coordinate axes perpendicular to each other are attached as necessary, and directions indicated by arrows are each set as the + direction, and directions opposite from the + direction are each set as the-direction. The +Z direction is also referred to as an upward side, and the −Z direction is also referred to as a downward side. In the present specification, a first direction represents a direction along the X-axis, and a second direction intersecting the first direction represents a direction along the Y-axis. In the following individual drawings, the scales of each layer and each member differ from those of actual layers and members in order to make each layer or each member have a recognizable size.

[0021] In addition, the plane including the X-axis and the Y-axis is also referred to as a XY plane, and a view of the XY plane as viewed from the +Z direction is also referred to as plan view or a planar view. Further, for example, with respect to a substrate, the expression “on the substrate” indicates any one of a case in which an object is disposed on the substrate in a contact manner, a case in which an object is indirectly disposed on the substrate with another structure being interposed therebetween, or a case in which a portion of an object is disposed on the substrate in a contact manner and a portion of the object is disposed with another structure being interposed therebetween.

First Embodiment

[0022] As an example of an electro-optical device, the present embodiment describes an active-drive type liquid crystal device **100** including a thin film transistor (hereinafter referred to as a “TFT”) serving as a transistor and provided for each pixel. The liquid crystal device **100** is favorably used as an optical modulation element (liquid-crystal light valve) of a projection-type display device serving as an electronic apparatus that will be described later, for example.

[0023] FIG. **1** is a schematic plan view illustrating a configuration of a liquid crystal device according to the present embodiment. FIG. **2** is a schematic cross-sectional view illustrating a structure of the liquid crystal device in a line segment A-A′ in FIG. **1**.

[0024] As illustrated in FIGS. **1** and **2**, the liquid crystal device **100** serving as an electro-optical device according to the present embodiment includes an element substrate **10**, a counter substrate **20** disposed to be opposed to the element substrate **10**, and a liquid crystal layer **5** including a liquid crystal interposed between the element substrate **10** and the counter substrate **20**.

[0025] A substrate such as a glass substrate or a quartz substrate is used for a substrate **10***a* of the element substrate **10**, for example. A transparent substrate such as a glass substrate or a quartz substrate is used for a substrate **20***a* of the counter substrate **20**, for example.

[0026] The element substrate **10** is larger than the counter substrate **20** in plan view. The element substrate **10** and the counter substrate **20** are bonded together through a sealing material **6** disposed along the outer edge of the counter substrate **20**. A liquid crystal having positive or negative dielectric anisotropy is incorporated in a space between the element substrate **10** and the counter substrate **20** to provide the liquid crystal layer **5**. In the following description, the element substrate **10** and the counter substrate **20** are also referred to as a pair of substrates.

[0027] An adhesive such as thermosetting, light curable, electron-beam curable epoxy resin is used for the sealing material **6**, for example. A spacer (not illustrated) is mixed with the sealing material **6** in order to maintain a constant distance between a pair of substrates.

[0028] A display region E including a plurality of pixels P arrayed in a matrix manner is provided at the inside of the sealing material **6**. A peripheral area F is provided at the outside of the display region E. A partition portion **23** is provided in the peripheral area F and between the sealing material **6** and the display region E to surround the display region E. Metal or metallic oxide having a light shielding property is used for the partition portion **23**.

[0029] Although not illustrated in the drawings, a dummy region is provided around the display region E, and a light shielding portion is provided in the display region E. The dummy region does not contribute to displaying performed by the liquid crystal device **100**. The light shielding portion is a so-called black matrix, and is provided in the counter substrate **20**.

[0030] A terminal portion in which a plurality of external coupling terminal **43** are arrayed is provided at the element substrate **10**. A data-line drive circuit **47** is provided between the first side along the terminal portion and the sealing material **6**. In addition, an inspection circuit **41** is provided between the display region E and the sealing material **6** along the second side opposed from the first side.

[0031] A scanning-line drive circuit **45** is provide between the display region E and the sealing material **6** along the third side and the fourth side that are perpendicular to the first side and are opposed to each other. A plurality of wiring lines **49** configured to couple two scanning-line drive circuits **45** are provided between the inspection circuit **41** and the sealing material **6** at the second side.

[0032] The data-line drive circuit **47** and the wiring lines **49** coupled to the scanning-line drive circuit **45** are electrically coupled to the plurality of external coupling terminals **43** arrayed along the first side. The arrangement of the inspection circuit **41** is not limited to that described above, and the inspection circuit **41** may be provided between the display region E and the sealing material **6** along the data-line drive circuit **47**.

[0033] As illustrated in FIG. **2**, a light-transmitting pixel electrode **11**, a TFT **30** serving as a switching element, the wiring lines **49**, and an alignment film **12** configured to cover these items are provided for each pixel P at the front surface, at the liquid crystal layer **5** side, of the substrate **10***a*. The TFT **30** and the pixel electrode **11** are constituent elements of the pixel P. The pixel electrode **11** is provided to correspond to the TFT **30**. The element substrate **10** includes the substrate **10***a*, the pixel electrode **11** provided on the substrate **10***a*, the TFT **30**, the wiring lines **49**, and the alignment film **12**.

[0034] Light L is incident on the liquid crystal device **100** from the counter substrate **20** side. The light L comes from a laser light source, for example. An incidence direction in which the light L is incident on the liquid crystal device **100** is not limited to the counter substrate **20** side, and the light L may be incident from the element substrate **10** side. In addition, the liquid crystal device **100** may be configured to include a light collecting unit such as a microlens configured to collect the incident light L for each pixel P.

[0035] At the front surface, at the liquid crystal layer **5** side, of the substrate **20***a*, there are provided the partition portion **23**, an insulating layer **25** covering the partition portion **23** and formed into a film, a counter electrode **21** provided to cover the insulating layer **25**, and an alignment film **22** provided to cover the counter electrode **21**. The liquid crystal device **100** is configured such that a common electrode is disposed at the counter substrate **20** side as the counter electrode **21**. However, the liquid crystal device **100** is not limited to this configuration.

[0036] As illustrated in FIG. **1**, the scanning-line drive circuit **45** and the inspection circuit **41** overlap the partition portion **23** in plan view. The partition portion **23** functions as a light shielding portion. Specifically, the partition portion **23** blocks light such that the light L described above incident from the counter substrate **20** side is not incident on a peripheral circuit such as the scanning-line drive circuit **45**. That is, the partition portion **23** has a function of preventing a malfunction of peripheral circuits. In addition, the partition portion **23** also curbs incidence of unnecessary stray light on the display region E. This makes it possible to curb a reduction in the contrast of the liquid crystal device **100**.

[0037] With reference back to FIG. **2**, the insulating layer **25** is configured to cover the partition portion **23** and is provided to flatten the front surface of the liquid crystal layer **5** side. The insulating layer **25** is made of an inorganic material such as silicon oxide having light transmittance, for example.

[0038] The counter electrode **21** is configured to cover the insulating layer **25**, and is electrically coupled to an up-down conductive portion **7** provided at four corners of the counter substrate **20**. The up-down conductive portion **7** is electrically coupled to a common wiring line **18**, which will be described later, at the element substrate **10** side.

[0039] The pixel electrode **11** and the counter electrode **21** are configured with a transparent conductive film made of indium tin oxide (ITO), indium zinc oxide (IZO), or the like, for example. The alignment film **12** and the alignment film **22** are selected on the basis of optical design of the liquid crystal device **100**. The material of the alignment films **12** and **22** includes an inorganic alignment film made of silicon oxide or the like and an organic alignment film made of polyimide or the like.

[0040] The optical design of a normally white mode or normally black mode is employed for the liquid crystal device **100**. In the normally white mode, the transmittance of a pixel P when no voltage is applied is larger than the transmittance when a voltage is applied. In the normally black mode, the transmittance of a pixel P when no voltage is applied is smaller than the transmittance when a voltage is applied. In the liquid crystal device **100**, a polarizing element is disposed at each of the side on which the light L is incident and the side from which the light L is emitted, depending on the optical design.

[0041] The present embodiment describes an example in which the inorganic alignment film described above is used as the alignment films **12** and **22**, a liquid crystal having negative dielectric anisotropy is used, and the optical design of the normally black mode is employed.

[0042] FIG. **3** is an equivalent circuit diagram illustrating an electrical configuration of the liquid crystal apparatus.

[0043] As illustrated in FIG. **3**, the liquid crystal device **100** includes a scanning line **13**, a data line **16**, and a common wiring line **18** provided on the substrate **10***a* of the element substrate **10**. The scanning line **13** extends along the X-axis. The data line **16** and the common wiring line **18** extend along the Y-axis. The common wiring line **18** is not limited to that extending along the Y-axis. The common wiring line **18** is an example of a first wiring line of the present disclosure.

[0044] A region separated by the scanning line **13** extending along the X-axis and the data line **16** extending along the Y-axis is a pixel P. The pixel P includes the pixel electrode **11**, the TFT **30**, a first capacitance element **50**, and a second capacitance element **60**. The first capacitance element **50** and the second capacitance element **60** are provided to correspond to the pixel electrode **11**.

[0045] The scanning line **13** is electrically coupled to the gate of the TFT **30**. The data line **16** is electrically coupled to the source of the TFT **30**. The scanning line **13** has a function of collectively controlling turning on and off the TFT **30** provided in the same row. The pixel electrode **11** is electrically coupled to the drain of the TFT **30**.

[0046] The data line **16** is electrically coupled to the data-line drive circuit **47**, and supplies the pixel P with image signals D**1**, D**2**, . . . , and Dn supplied from the data-line drive circuit **47**. The scanning line **13** is electrically coupled to the scanning-line drive circuit **45**, and supplies each pixel P with scanning signals SC**1**, SC**2**, . . . , and SCm supplied from the scanning-line drive circuit **45**.

[0047] The image signal D**1** to the image signal Dn supplied from the data-line drive circuit **47** to the data line **16** may be supplied in a line-sequential manner in this order, or may be supplied, on a group-by-group basis, to a plurality of data lines **16** adjacent to each other. The scanning-line drive circuit **45** supplies the scanning line **13** with the scanning signal SC**1** to the scanning signal SCm at a predetermined timing in a pulse form in a line-sequential manner.

[0048] When the scanning signal SC**1** is input to the TFT **30**, the TFT **30** is brought into the ON state for a predetermined period of time. Thereby, the image signal D**1** supplied from the data line **16** is written in the pixel electrode **11** at a predetermined timing. In addition, the image signal DI at a predetermined level written in the liquid crystal layer **5** through the pixel electrode **11** is held for a certain period of time between the pixel electrode **11** and the counter electrode **21** disposed to be opposed to each other with the liquid crystal layer **5** being interposed between them.

[0049] In order to prevent the held image signal D**1** from leaking, the first capacitance element **50** and the second capacitance element **60** are electrically coupled in parallel with respect to a liquid crystal capacitor provided between the pixel electrode **11** and the counter electrode **21**. One end of the first capacitance element **50** is electrically coupled to the drain of the TFT **30** and the pixel electrode **11**. The other end of the first capacitance element **50** is electrically coupled to the common wiring line **18** to which a constant potential is applied. Similarly to the first capacitance element **50**, the second capacitance element **60** is electrically coupled to the common wiring line **18**.

[0050] Here, although not illustrated in the drawings, the inspection circuit **41** is coupled to the data line **16**. During the process of manufacturing the liquid crystal device **100**, this makes it possible to detect the image signals D**1**, D**2**, . . . , and Dn to confirm operational malfunction or the like of the liquid crystal device **100**.

[0051] FIG. **4** is a schematic plan view illustrating arrangement of pixels.

[0052] As illustrated in FIG. **4**, the pixel electrode **11** has a substantially square shape, and there are a plurality of pixel electrodes **11** provided in a matrix manner to correspond to the arrangement of pixels P. Between adjacent pixel electrodes **11**, the scanning line **13** is provided along the X-axis, and the data line **16** and the common wiring line **18** are provided along the Y-axis. In FIG. **4**, illustration is made such that a region E**1** that is a portion of the display region E in FIG. **1** is enlarged.

[0053] In the display region E, a plurality of pixels P are divided in plan view to provide a light shielding region SD. The light shielding region SD includes a straight-shaped portion including the scanning line **13**, and a straight-shaped portion including the data line **16** and the common wiring line **18**, and has a lattice shape as shown by a dashed line. An electrically conductive member having a light shielding property is used for the functional layers and wiring lines such as the scanning line **13** and the data line **16**, and hence, the light shielding region SD is a so-called non-open area. That is, in the display region E, a region except for the light shielding region SD is an opening area.

[0054] A second contact hole CNT**12** is provided in an end portion, at the −Y direction, of the light shielding region SD so as to correspond to the pixel electrode **11**. The pixel electrode **11** and a third relay electrode **83** that will be described later are electrically coupled to each other through the second contact hole CNT**12**. That is, in the present embodiment, the third relay electrode **83** corresponds to a third relay electrode of a typical electro-optical device.

[0055] In the liquid crystal device **100**, the second contact hole CNT**12** is provided in the light shielding region SD. That is, the scanning line **13** and the common wiring line **18** are not recessed in plan view, and hence, it is possible to further reduce the width of these wiring lines, which makes it possible to easily improve the aperture ratio. Detailed configuration of the element substrate **10** including the third relay electrode **83** and the like will be described later.

[0056] FIG. **5** is a cross-sectional view illustrating a structure of an element substrate in a line segment β**1**-β**2** in FIG. **4**. FIG. **6** is a cross-sectional view illustrating a structure of an element substrate in a line segment α**1**-α**2** in FIG. **4**. FIG. **7** is a cross-sectional view illustrating a structure of an element substrate in a line segment γ**1**-γ**2** in FIG. **4**.

[0057] As illustrated in FIGS. **5**, **6**, and **7**, the element substrate **10** of the liquid crystal device **100** includes the substrate **10***a*, the second capacitance element **60**, the scanning line **13**, the TFT **30** including a semiconductor layer **31** and a gate electrode **32**, the first capacitance element **50**, the data line **16**, the common wiring line **18**, the third relay electrode **83**, the pixel electrode **11**, and a plurality of interlayer insulating layers that will be described later. The third relay electrode **83** is an example of a relay layer of the present disclosure.

[0058] The element substrate **10** has a configuration in which a plurality of functional layers are stacked on the substrate **10***a* serving as a base. Specifically, on the substrate **10***a*, there are stacked, in this order, a first conductive layer including a fourth capacitance electrode **62** of the second capacitance element **60**, a second conductive layer including a third capacitance electrode **61** of the second capacitance element **60**, a third conductive layer including the scanning line **13**, a fourth conductive layer including the semiconductor layer **31** of the TFT **30** and the gate electrode **32** of the TFT **30**, a fifth conductive layer including a second capacitance electrode **52** of the first capacitance element **50**, a sixth conductive layer including a first capacitance electrode **51** of the first capacitance element **50**, a seventh conductive layer including the data line **16**, an eighth conductive layer including the common wiring line **18**, and the pixel electrode **11**.

[0059] A second dielectric layer **63** is provided between the fourth capacitance electrode **62** of the second capacitance element **60**, which is the first conductive layer, and the third capacitance electrode **61** of the second capacitance element **60**, which is the second conductive layer. A first interlayer insulating layer **71** is provided between the second conductive layer and the scanning line **13** which is the third conductive layer. A second interlayer insulating layer **72** is provided between the third conductive layer and the semiconductor layer **31** of the TFT **30**. A gate insulating layer **33** is provided between the semiconductor layer **31** and the gate electrode **32**. A third interlayer insulating layer **73** serving as a second insulating layer is provided between the TFT **30**, which is the fourth conductive layer, and the second capacitance electrode **52** of the first capacitance element **50**, which is the fifth conductive layer.

[0060] A first dielectric layer **53** is provided between the second capacitance electrode **52** of the first capacitance element **50**, which is the fifth conductive layer, and the first capacitance electrode **51** of the first capacitance element **50**, which is the sixth conductive layer. A fourth interlayer insulating layer **74** serving as a first insulating layer is provided between the first capacitance electrode **51**, which is the sixth conductive layer, and the data line **16**, which is the seventh conductive layer. A fifth interlayer insulating layer **75** is provided between the data line **16**, which is the seventh conductive layer, and the common wiring line **18**, which is the eighth conductive layer. A sixth interlayer insulating layer **76** is provided between the common wiring line **18**, which is the eighth conductive layer, and the pixel electrode **11**.

[0061] The materials of these interlayer insulating layers include silicon oxide (None-doped Silicate Glass: NSG) or silicon nitride or the like, for example. In the present embodiment, silicon oxide is employed.

[0062] The second capacitance element **60** is provided on the substrate **10***a* and also inside of a trench that is not illustrated in the drawing. The second capacitance element **60** is configured such that the fourth capacitance electrode **62**, the second dielectric layer **63**, and the third capacitance electrode **61** are stacked in the order from the substrate **10***a* side. The materials of the third capacitance electrode **61** and the fourth capacitance electrode **62** include electrically conductive polysilicon, for example. A dielectric material is used for the material of the second dielectric layer **63**. For example, the dielectric material includes silicon nitride, silicon oxide, hafnium oxide, aluminum oxide, tantalum oxide, or the like, and these layers are used as a single layer or as a combination of them.

[0063] The scanning line **13** also functions as a light shielding layer, and is provided on the first interlayer insulating layer **71**. A known material having a light shielding property and electrically conductive property is used for the scanning line **13**. The scanning line **13** has a function of blocking the light L incident on the semiconductor layer **31** mainly from below. In the present embodiment, tungsten silicide is used as the material of the scanning line **13**.

[0064] The TFT **30** includes the semiconductor layer **31** provided on the second interlayer insulating layer **72**, the gate insulating layer **33**, and the gate electrode **32** provided on the third interlayer insulating layer **73**. The gate electrode **32** is electrically coupled to the scanning line **13**.

[0065] The semiconductor layer **31** of the TFT **30** is provided with a lightly doped drain (LDD) structure. The semiconductor layer **31** is made of conductive polysilicon, for example. The gate electrode **32** is made of conductive polysilicon, for example. The semiconductor layer **31** extends along the Y-axis.

[0066] The first capacitance element **50** is provided on the third interlayer insulating layer **73**. The first capacitance element **50** is configured such that the second capacitance electrode **52**, the first dielectric layer **53**, and the first capacitance electrode **51** are stacked in the order from the substrate **10***a* side. The materials of the first capacitance electrode **51** and the second capacitance electrode **52** include electrically conductive polysilicon, for example. A dielectric material similar to that of the second capacitance element **60** is used as the material of the first dielectric layer **53**.

[0067] The data line **16** is provided on the fourth interlayer insulating layer **74**, and extends along the Y-axis. There is no particular limitation as to the material of the data line **16** as long as the material is a low-resistance wiring line material having electrical conductivity, and the material of the data line **16** includes a metal such as aluminum or titanium, or a metallic compound of these metal, for example. The data line **16** is electrically coupled to a source-drain region of the semiconductor layer **31**.

[0068] The common wiring line **18** and the third relay electrode **83** are provided in the same layer at the fifth interlayer insulating layer **75**. The common wiring line **18** and the third relay electrode **83** are electrically coupled to the counter electrode **21** described above, and are supplied with a common potential. The materials of the common wiring line **18** and the third relay electrode **83** include a metal such as aluminum or titanium, or a metallic compound of these metals, for example. In the liquid crystal device **100**, a stacked structure is employed for the common wiring line **18** and the third relay electrode **83**.

[0069] Specifically, the common wiring line **18** includes a first layer **18***a* having light reflectance, a second layer **18***b* stacked on the first layer **18***a* and having light reflectance lower than the first layer **18***a*, and a third layer **18***c* staked on the second layer **18***b* and having light reflectance higher than the second layer **18***b*. The third relay electrode **83** includes a first layer **83***a* having light reflectance, a second layer **83***b* stacked on the first layer **83***a* and having light reflectance lower than the first layer **83***a*, and a third layer **83***c* stacked on the second layer **83***b* and having light reflectance higher than the second layer **83***b*.

[0070] The first layers **18***a* and **83***a* contain aluminum, the second layers **18***b* and **83***b* contain titanium nitride., and the third layers **18***c* and **83***c* contain titanium. Specifically, in the present embodiment, aluminum is adopted for the first layers **18***a* and **83***a*, titanium nitride is adopted for the second layers **18***b* and **83***b*, and titanium is adopted for the third layers **18***c* and **83***c*. The titanium nitride used in the second layers **18***b* and **83***b* exerts effect of curbing electrolytic corrosion and oxidation of the aluminum in the first layers **18***a* and **83***a* and protecting the first layers **18***a* and **83***a* from a stripping solution used in a patterning process during manufacturing. On the other hand, aluminum has favorable electrical conductivity, has light reflectance higher than titanium nitride, and exerts effect of reflecting light L by not providing the second layers **18**_b_ and **83**_b_ in a region that does not require the above-mentioned effect to curb generation of heat within the element substrate **10**. In addition, titanium has light reflectance higher than titanium nitride, and exerts effect of reflecting light L in a region that requires the above-mentioned effect the second layers **18**_b_ and **83**_b_ by disposing titanium on the second layers **18**_b_ and **83**_b_ in an overlapping manner to efficiently curb heat generation inside the element substrate **10**.

[0071] Here, when the films are formed using a sputtering method, the light reflectance of aluminum is approximately 90% in a wavelength of 550 nm in a visible range, the light reflectance of titanium nitride is approximately 25%, and the light reflectance of titanium is approximately 55%, for example. The light reflectance of such a base material can be measured using a spectro photometer or the like. In the present specification, the “having light reflectance” means that the light reflectance described above is approximately 50% or more. With this configuration, by actively reflecting light to prevent light absorption, it is possible to curb heat generation inside the liquid crystal device **100**.

[0072] In the present embodiment, the areas of the third layers **18**_c_ and **83**_c_ are equal to the areas of the second layers **18**_b_ and **83**_b_ in plan view. When the element substrate **10** is seen in plan view, the areas of the third layers **18**_c_ and **83**_c_ that cover the second layers **18**_b_ and **83**_b_ are smaller than the areas of the corresponding first layers **18**_a_ and **83**_a_. That is, the second layers **18**_b_ and **83**_b_ and the third layers **18**_c_ and **83**_c_ are stacked at a portion of the region of the corresponding first layers **18**_a_ and **83**_a_, rather than being stacked in the entire regions of the first layers **18**_a_ and **83**_a_. Details of the common wiring line **18** and the third relay electrode **83** will be described later.

[0073] The pixel electrode **11** is provided on the sixth interlayer insulating layer **76**. Although not illustrated in the drawings, the alignment film **12** is provided to cover the pixel electrode **11**. The alignment film **12** of the element substrate **10** and the alignment film **22** and the counter substrate **20** described above are each configured with a collective body of columns obtained by depositing an inorganic material such as silicon oxide from a predetermined direction such as an oblique direction and growing the inorganic material into a column shape.

[0074] As illustrated in FIG. **5**, the second capacitance electrode **52** of the first capacitance element **50** is electrically coupled to a fifth relay electrode **85** disposed at the fourth conductive layer, and the fifth relay electrode **85** is electrically coupled to the fourth capacitance electrode **62** of the second capacitance element **60**.

[0075] The second layer **18**_b_ and the third layer **18**_c_ are not provided in a range of the line segment β**1**-β**2** of the common wiring line **18**. The light L incident from above is reflected by the first layer **18**_a_ in a region where the second layer **18**_b_ and the third layer **18**_c_ are not stacked. Thus, in this region described above, it is possible to curb generation of heat due to absorption of the light L.

[0076] As illustrated in FIG. **6**, the common wiring line **18** and the first capacitance electrode **51** of the first capacitance element **50** are electrically coupled through a first relay electrode **81** disposed in the seventh conductive layer. Furthermore, the first relay electrode **81** is electrically coupled to a second relay electrode **82** disposed in the fourth conductive layer.

[0077] The third relay electrode **83** is electrically coupled to a fourth relay electrode **84** disposed in the seventh conductive layer. The fourth relay electrode **84** is electrically coupled to the second capacitance electrode **52** of the first capacitance element **50**.

[0078] The third relay electrode **83** is formed to extend into a first contact hole CNT**11** in a region overlapping the first contact hole CNT**11**. The first layer **83**_a_, the second layer **83**_b_, and the third layer **83**_c_ are provided along the inner surface of the first contact hole CNT**11**. More specifically, the first layer **83**_a_ is routed along the inner surface of the first contact hole CNT**11** and is electrically coupled to the fourth relay electrode **84** located on the bottom surface of the first contact hole CNT**11**. The second layer **83**_b_ is disposed along the surface of the first layer **83**_a_, and the third layer **83**_c_ is disposed along the surface of the second layer **83**_b_, and thus disposed along the inner surface of the first contact hole CNT**11**.

[0079] The aluminum that constitutes the first layer **83***a* of the third relay electrode **83** is easily etched using a stripping solution for a resist film used in a patterning process. The first layer **83***a* tends to have poor adhesion inside the contact hole and to have a thin film thickness. For this reason, when the first layer **83***a* is etched, there is a concern that the reliability of electrical coupling will be reduced.
[0080] In contrast, in the case of the present embodiment, the first layer **83***a* is covered and protected by the second layer **83***b* and the third layer **83***c* in the process of patterning the third relay electrode **83**, and thus etching of the first layer **83***a* is curbed and the reliability of electrical coupling is improved.
[0081] As illustrated in FIGS. **6** and **7**, the first capacitance electrode **51** of the first capacitance element **50** and the third capacitance electrode **61** of the second capacitance element **60** are electrically coupled to the common wiring line **18** through the first relay electrode **81** disposed in the seventh conductive layer and the second relay electrode **82** disposed in the fourth conductive layer.
[0082] The common wiring line **18** is provided to extend into a third contact hole CNT**13** in a region overlapping the third contact hole CNT**13**. The first layer **18***a*, the second layer **18***b* and the third layer **18***c* are provided along the inner surface of the third contact hole CNT**13**. More specifically, the first layer **18***a* is routed along the inner surface of the third contact hole CNT**13** and is electrically coupled to the first relay electrode **81** located on the bottom surface of the third contact hole CNT**13**. The second layer **18***b* is disposed along the surface of the first layer **18***a*, and the third layer **18***c* is disposed along the surface of the second layer **18***b*, and thus is disposed along the inner surface of the third contact hole CNT**13**.
[0083] Aluminum constituting the first layer **18***a* of the common wiring line **18** is easily etched using a stripping solution for a resist film used in a patterning process. The first layer **18***a* tends to have poor adhesion inside the contact hole and to have a thin film thickness. For this reason, when the first layer **18***a* is etched, there is a concern that the reliability of electrical coupling will be reduced.
[0084] In contrast, in the case of the present embodiment, the first layer **18***a* is covered and protected by the second layer **18***b* and the third layer **18***c* in the process of patterning the common wiring line **18**, and thus etching of the first layer **18***a* is curbed and the reliability of electrical coupling is improved.
[0085] As illustrated in FIG. **7**, the second relay electrode **82** is electrically coupled to the third capacitance electrode **61** of the second capacitance element **60**. Specifically, although not illustrated in the drawings, the third capacitance electrode **61** includes a sticking-out portion **61***a* that sticks out in plan view. The second relay electrode **82** is electrically coupled to this sticking-out portion **61***a*. The second capacitance electrode **52** of the first capacitance element **50** and the fourth capacitance electrode **62** of the second capacitance element **60** are electrically coupled to the pixel electrode **11** and a drain **31***d* of the TFT **30**. Instead of the first capacitance element **50**, it may be possible to use a relay electrode used to electrically couple the fourth capacitance electrode **62** of the second capacitance element **60** and the drain **31***d* of the TFT **30**.
[0086] The pixel electrode **11** is electrically coupled to the third relay electrode **83** disposed in the eighth conductive layer. In particular, the pixel electrode **11** is electrically coupled to the third relay electrode **83** through the second contact hole CNT**12**. In the third relay electrode **83**, the second layer **83***b* and the third layer **83***c* are covered on the first layer **83***a* in a region including the region where the third relay electrode **83** overlaps the second contact hole CNT**12**. In the case of the present embodiment, the second layer **83***b* covers the first layer **83***a*, and the third layer **83***c* covers the second layer **83***b*. That is, the second layer **83***b* and the third layer **83***c* are interposed between the pixel electrode **11** and the first layer **83***a*.
[0087] Here, when aluminum and ITO are electrically coupled directly to each other, electrolytic corrosion is more likely to occur between them. This may lead to breakage of aluminum lines or

insulation due to generation of alumina or the like. In contrast, in the present embodiment, titanium and ITO are coupled, which curbs the occurrence of electrolytic corrosion in the region including the second contact hole CNT**12**, improving the reliability of electrical coupling. Only the second layer **83***b* may be covered on the first layer **83***a*. Even in this case, the coupling between titanium nitride and ITO curbs the occurrence of electrolytic corrosion, improving the reliability of electrical coupling.

[0088] The signal wiring lines such as scanning line **13** and the common wiring line **18** that constitute the functional layers described above and the electrodes such as the TFT **30** and the first relay electrode **81** are provided in the light shielding region SD that separates a plurality of pixels P in plan view.

[0089] In manufacturing of the element substrate **10**, it may be possible to employ a known method applied to known semiconductor processes, which includes a low pressure chemical vapor deposition (CVD) method, an atmospheric pressure CVD method, a plasma CVD method, a photolithography method, a sputtering method, an etching method, a chemical mechanical planarization (CMP) method, and the like.

[0090] FIG. **8** is a plan view illustrating configurations of common wiring lines and third relay electrodes.

[0091] As illustrated in FIG. **8**, common wiring lines **18** are disposed along the Y-axis in a stripe manner. The element substrate **10** includes a common wiring line **18** serving as a first wiring line at the center in FIG. **8**, and a common wiring line **18** serving as a second wiring line adjacent in a direction along the X-axis. The third relay electrode **83** is provided between common wiring lines **18** adjacent to each other in plan view. Thereby, the common wiring lines **18** and the third relay electrode **83** can be disposed in a state of being separated from each other.

[0092] FIG. **8** illustrates a region corresponding to FIG. **4**. Additionally, in the common wiring line **18**, the region to which the stacked structure configured with the first layer **18***a*, the second layer **18***b*, and the third layer **18***c* is applied is indicated by hatching. Similarly, also in the third relay electrode **83**, the region to which the stacked structure configured with the first layer **83***a*, the second layer **83***b*, and the third layer **83***c* is applied is indicated by hatching. Furthermore, the boundary between a main body portion **18**M and a protruding portion **18**P, which will be described later, is shown by a dashed line.

[0093] The common wiring line **18** includes the main body portion **18**M and the protruding portion **18**P. The main body portion **18**M extends in a direction along the Y-axis. The protruding portion **18**P protrudes from the main body portion **18**M in the −X direction. The main body portion **18**M and the protruding portion **18**P are examples of a second main body portion and a second protruding portion of the present disclosure.

[0094] The protruding portion **18**P is configured such that the first layer **18***a* and the second layer **18***b* are provided in a region including a region that overlaps the third contact hole CNT**13** in plan view. Specifically, the protruding portion **18**P is configured such that the third contact hole CNT**13** is disposed in a region including a region that overlaps, in plan view, the vicinity of the end portion at the −X direction of the protruding portion **18**P.

[0095] The common wiring line **18** is electrically coupled to the first capacitance electrode **51** of the first capacitance element **50** described above through the third contact hole CNT**13**.

[0096] The second layer **18***b* and the third layer **18***c* are not provided at at least a portion of the main body portion **18**M. That is, in the present embodiment, the second layer **18***b* and the third layer **18***c* are not provided over the entire region of the main body portion **18**M in plan view. In addition, the protruding portion **18**P is configured such that the second layer **18***b* and the third layer **18***c* are not provided at a portion in the +X direction which is the common wiring line **18** side.

[0097] The arrangement described above is preferable when the distance between adjacent common wiring lines **18** along the X-axis is relatively wide. That is, since the above-mentioned distance is relatively wide, the functions of the second layer **18***b* and the third layer **18***c* are sufficiently exerted around the third contact hole CNT**13** even when the areas of the second layer **18***b* and the third layer **18***c* are relatively small. Thereby, the surplus second layer **18***b* and third layer **18***c* around the third contact hole CNT**13** are reduced. For this reason, the region where the first layer **18***a* is exposed is increased, further curbing heat generation inside the element substrate **10**, while improving the reliability of electrical coupling around the third contact hole CNT**13**.

[0098] The third relay electrode **83** includes a main body portion **83**M and a protruding portion **83**P. The main body portion **83**M has an island shape, and extends in a direction along the X-axis. The protruding portion **83**P protrudes from the main body portion **83**M in the +Y direction. The main body portion **83**M and the protruding portion **83**P are examples of a first main body portion and a first protruding portion of the present disclosure.

[0099] The third relay electrode **83** is configured such that the first layer **83***a*, the second layer **83***b*, and the third layer **83***c* are provided in a region including a region that overlaps the first contact hole CNT**11** and the second contact hole CNT**12** in plan view. Specifically, in the main body portion **83**M, the first contact hole CNT**11** is disposed in a region including a region that overlaps the end portion at the −X direction and the vicinity of this end portion in plan view. The second contact hole CNT**12** is disposed in a region including a region that overlaps the protruding portion **83**P in plan view. In the third relay electrode **83**, the entire region of the first layer **83***a* is covered with the second layer **83***b* and the third layer **83***c* in plan view.

[0100] The protruding portion **83**P is electrically coupled to the pixel electrode **11** described above through the second contact hole CNT**12**. The main body portion **83**M is electrically coupled to the TFT **30** described above through the first contact hole CNT**11**.

[0101] The first layers **18***a* and **83***a*, the second layers **18***b* and **83***b*, and the third layers **18***c* and **83***c* can be formed by a known technique such as sputtering. In the direction along the Z-axis, the first layers **18***a* and **83***a* have a thickness of, for example, 0.35 μm, the second layers **18***b* and **83***b* have a thickness of, for example, 0.13 μm, and the third layers **18***c* and **83***c* have a thickness of, for example, 0.02 μm.

[0102] According to the liquid crystal device **100** of the present embodiment, the following effects can be obtained.

[0103] According to the liquid crystal device **100** of the present embodiment, an aperture ratio and the reliability of the electrical couplings in the first contact hole CNT**11**, the second contact hole CNT**12**, and the third contact hole CNT**13** are improved, and heat generation inside the element substrate **10** can be curbed. More specifically, the third relay electrode **83** and the pixel electrode **11** are electrically coupled by the second contact hole CNT**12** that is disposed in a region including the region that overlaps the protruding portion **83**P in plan view. This configuration makes it possible to reduce the width and the area of the main body portion **83**M in plan view, as compared with a case in which the second contact hole CNT**12** is disposed in a region including a region that overlaps the main body portion **83**M of the third relay electrode **83** in plan view. This makes it possible to improve the aperture ratio.

[0104] Since the protruding portions **18**P and **83**P have a stacked structure of aluminum, titanium nitride, and titanium, the first contact hole CNT**11** and the third contact hole CNT**13** also have a stacked structure in which aluminum, titanium nitride, and titanium are stacked. For this reason, when the third relay electrode **83** and the common wiring line **18** are patterned, the first layers **18***a* and **83***a* are protected by the second layers **18***b* and **83***b* and the third layers **18***c* and **83***c*, and etching of the first layers **18***a* and **83***a* is curbed. This makes it possible to improve the reliability of electrical coupling through the first contact hole CNT**11** and the third contact hole CNT**13**.

[0105] The third layer **18***c* covering the outermost surface of the common wiring line **18** easily reflects light L, thereby curbing heat generation inside the element substrate **10** due to absorption of the light L in the common wiring line **18**. Furthermore, the second layer **18***b* and the third layer **18***c* are not provided in at least a portion of the region of the common wiring line **18**, that is, portions of the regions of the main body portion **18**M and the protruding portion **18**P. The first layer **18***a* easily reflects light L compared to the second layer **18***b* and the third layer **18***c*, thereby curbing heat generation inside the element substrate **10** due to absorption of the light L. Thus, it is possible to curb heat generation in the liquid crystal device **100**.

[0106] The second contact hole CNT**12** is provided in a region that overlaps the protruding portion **83**P of the third relay electrode **83** having the stacked structure. For this reason, since the second layer **83***b* and the third layer **83***c* are interposed between the first layer **83***a* and the pixel electrode **11**, it is possible to curb the occurrence of electrolytic corrosion and improve the reliability of electrical coupling.

[0107] As described above, according to the present embodiment, it is possible to provide the liquid crystal device **100** that improves an aperture ratio and the reliability of electrical coupling between the first contact hole CNT**11** and the third contact hole CNT**13**, and curbs heat generation inside the element substrate **10**.

Second Embodiment

[0108] A liquid crystal device serving as an electro-optical device according to the present embodiment will be described with reference to FIGS. **9** to **11**. A liquid crystal device according to a second embodiment is configured such that a region where a second layer **18***b* of a common wiring line **18** is provided is modified from the liquid crystal device **100** according to the embodiment described above. Thus, the same reference characters are used for the same constituent portions as those in the first embodiment, and explanation thereof will not be repeated.

[0109] FIG. **9** is a plan view illustrating a mode of a third relay electrode according to the second embodiment. FIG. **9** illustrates a region corresponding to FIG. **4**, as in FIG. **8**. Additionally, in the common wiring line **18**, a region to which a stacked structure of a first layer **18***a*, the second layer **18***b*, and a third layer **18***c* is applied is indicated by hatching. Similarly, in a third relay electrode **83**, a region to which a stacked structure of a first layer **83***a*, a second layer **83***b*, and a third layer **83***c* is applied is indicated by hatching.

[0110] As illustrated in FIG. **9**, the second layer **18***b* and the third layer **18***c* are not provided in at least a portion of a main body portion **18**M. That is, in the present embodiment, the second layer **18***b* and the third layer **18***c* are provided along the X-axis in plan view in the entire region of a protruding portion **18**P and a portion of the region of the main body portion **18**M.

[0111] FIG. **10** is a cross-sectional view illustrating a structure of an element substrate in a line segment β**1**-β**2** in FIG. **4**. FIG. **11** is a cross-sectional view illustrating a structure of an element substrate in a line segment α**1**-α**2** in FIG. **4**.

[0112] More specifically, as illustrated in FIG. **10**, the second layer **18***b* and the third layer **18***c* are stacked at a portion of the first layer **18***a* in the cross-section of the line segment β**1**-β**2**. FIG. **10** illustrates a region corresponding to FIG. **5**. In addition, as illustrated in FIG. **11**, the entire region of the first layer **18***a* is covered with the second layer **18***b* and the third layer **18***c* in the cross-section of the line segment α**1**-α**2**. FIG. **11** illustrates a region corresponding to FIG. **6**.

[0113] The arrangement described above is preferable when the distance between adjacent common wiring lines **18** along the X-axis is relatively narrow. That is, since the above-mentioned distance is relatively narrow, the areas of the second layer **18***b* and the third layer **18***c* can be made relatively large around a third contact hole CNT**13**, and thus it is possible to exhibit a function of improving light reflectance by the third layer **18***c* while sufficiently exhibiting a function of curbing electrolytic corrosion and oxidation by the second layer **18***b*.

[0114] With the present embodiment, it is possible to obtain the effects similar to those of the embodiment described above.

[0115] In this embodiment, as shown in FIG. **11**, the third relay electrode **83** is configured such that the first layer **83***a* and the second layer **83***b* are provided along the inner surface of a first contact hole CNT**11**, and the third layer **83***c* fills the first contact hole CNT**11**. Thereby, the third layer **83***c* can flatten a recess caused by the first contact hole CNT**11**. Thus, the film formation process is facilitated when manufacturing the layer above the first contact hole CNT**11**.

[0116] Furthermore, the common wiring line **18** is configured such that the first layer **18***a* and the second layer **18***b* are provided along the inner surface of a third contact hole CNT**13**, and the third layer **18***c* fills the third contact hole CNT**13**. Thereby, the third layer **18***c* can flatten a recess caused by the third contact hole CNT**13**. Thus, the film formation process is facilitated when manufacturing the layer above the third contact hole CNT**13**.

Third Embodiment

[0117] The present embodiment describes, as an example, a projection-type display device **1000** as an electronic apparatus. The liquid crystal device **100** serving as the electro-optical device according to the embodiments described above is mounted at the projection-type display device **1000** according to the present embodiment. The projection-type display device **1000** is a liquid crystal projector.

[0118] FIG. **12** is a schematic view illustrating a configuration of a projection-type display device according to a third embodiment.

[0119] As illustrated in FIG. **12**, the projection-type display device **1000** includes: a polarized-light illumination device **1100** disposed along a system optical axis LS; two dichroic mirrors **1104** and **1105** serving as a light splitting element; three reflecting mirrors **1106**, **1107**, and **1108**; five relay lenses **1201**, **1202**, **1203**, **1204**, and **1205**; transmissive-type liquid-crystal light valves **1210**, **1220**, and **1230** serving as three optical modulation elements; a cross dichroic prism **1206** serving as a light synthesizing element; and a projection lens **1207**.

[0120] The polarized-light illumination device **1100** generally includes a lamp unit **1101** being as a light source including a white light source such as an extra-high pressure mercury lamp or a halogen lamp, an integrator lens **1102**, and a polarization conversion element **1103**.

[0121] The dichroic mirror **1104** reflects red light (R) of a polarized light flux emitted from the polarized-light illumination device **1100**, and allows green light (G) and blue light (B) to pass through. The other dichroic mirror **1105** reflects the green light (G) passing through the dichroic mirror **1104**, and allows the blue light (B) to pass through.

[0122] The red light (R) reflected by the dichroic mirror **1104** is reflected by the reflection mirror **1106** and is then incident on the liquid-crystal light valve **1210** via the relay lens **1205**. The green light (G) reflected by the dichroic mirror **1105** is incident on the liquid-crystal light valve **1220** via the relay lens **1204**. The blue light (B) transmitted by the dichroic mirror **1105** is incident on the liquid crystal light valve **1230** via a light guide system including the three relay lenses **1201**, **1202**, and **1203** and the two reflection mirrors **1107** and **1108**.

[0123] The liquid-crystal light valves **1210**, **1220**, and **1230** are each disposed to be opposed to a corresponding incident surface of each type of color light of the cross dichroic prism **1206**. The color light that is incident on the liquid-crystal light valves **1210**, **1220**, and **1230** is modulated on the basis of image information (image signal), and is output toward the cross dichroic prism **1206**.

[0124] In the cross dichroic prism **1206**, four right-angle prisms are bonded together, and on inner surfaces of the prisms, a dielectric multilayer film configured to reflect the red light and a dielectric multilayer film configured to reflect the blue light are disposed in a cross shape. The three types of color light are synthesized by these dielectric multilayer films to obtain light representing a color image. The thus obtained light is projected onto a screen **1300** through the projection lens **1207** serving as a projection optical system, and an image is enlarged to be displayed.

[0125] The liquid crystal device **100** described above is applied to the liquid-crystal light valves **1210**. The same applies to the other liquid-crystal light valves **1220** and **1230**.

[0126] The projection-type display device **1000** as described above includes the liquid crystal device **100** according to the first embodiment. This makes it possible to improve the reliability of electrical coupling in a region including the first contact hole CNT**11**, the second contact hole CNT**12**, and the third contact hole CNT**13** of the element substrate **10**. Furthermore, it is possible to curb generation of heat within the element substrate **10**. It is possible to obtain the similar effects by employing the liquid crystal device according to the second embodiment as the liquid-crystal light valves **1210**, **1220**, and **1230**.

[0127] In addition to the projection-type display device **1000**, the liquid crystal device **100** may be mounted on various types of electronic apparatuses such as an electrical view finder (EVF), a mobile mini-projector, a head-up display, a smartphone, a mobile phone, a mobile computer, a digital camera, a digital video camera, a display, a vehicle-mounted unit, an audio unit, a light exposure device, or an illumination unit.

[0128] The technical scope of the present disclosure is not limited to the above-described embodiments, and various modifications can be made without departing from the spirit and gist of the present disclosure. In addition, one aspect of the present disclosure can be configured by appropriately combining the characteristic portions of the above-described embodiments.

Overview of Present Disclosure

[0129] An overview of the present disclosure is provided below as the appendices.

Appendix 1

[0130] An electro-optical device including: [0131] a pixel electrode; [0132] a first wiring line including a first layer, a second layer, and a third layer; [0133] a transistor at least partially overlapping the first wiring line in plan view; and [0134] a data line provided in a layer between the first wiring line and the transistor, at least a portion of the data line overlapping the first wiring line in plan view, and electrically coupled to the pixel electrode via the transistor, wherein [0135] the first layer has light reflectance and is disposed closer to the transistor than the second layer and the third layer, [0136] the second layer is provided between the first layer and the third layer and has light reflectance lower than the first layer, and [0137] the third layer has light reflectance higher than the second layer.

[0138] According to the electro-optical device having the configuration of Appendix 1, the third layer covering the outermost surface of the first wiring line easily reflects light, and thus it is possible to curb head generation due to absorption of light by the first wiring line. Thus, according to this configuration, it is possible to provide the electro-optical device in which heat generation is curbed.

Appendix 2

[0139] The electro-optical device according to Appendix 1, further including a relay layer including a first main body portion provided in the same layer as the first wiring line, extending along a first direction, and electrically coupled to the transistor via a first contact hole, and a first protruding portion protruding from the first main body portion in a second direction intersecting the first direction, and electrically coupled to the pixel electrode via a second contact hole, [0140] wherein the relay layer includes the first layer, the second layer, and the third layer.

[0141] According to the configuration of Appendix 2, it is possible to configure the relay layer that electrically couples the transistor and the pixel electrode with a stacked structure of the first layer, the second layer, and the third layer.

Appendix 3

[0142] The electro-optical device according to Appendix 2, wherein at least the first layer and the second layer of the relay layer are provided along an inner surface of the first contact hole.

[0143] According to the configuration of Appendix 3, the first layer provided in the second contact hole is covered with the second layer, and thus it is possible to curb etching of the first layer in the second contact hole during patterning of the relay layer.

Appendix 4

[0144] The electro-optical device according to Appendix 3, wherein the third layer of the relay layer flattens a recess caused by the second contact hole.

[0145] According to the configuration of Appendix 4, a film forming process is facilitated when manufacturing a layer above the second contact hole.

Appendix 5

[0146] The electro-optical device according to any one of Appendices 2 to 4, further including a

second wiring line adjacent to the first wiring line, [0147] wherein the relay layer is provided between the first wiring line and the second wiring line when seen in plan view.
[0148] According to the configuration of Appendix 5, the first wiring line and the second wiring line can be disposed in a state of being separated from the relay layer.
Appendix 6
[0149] The electro-optical device according to any one of Appendices 2 to 4, wherein the first wiring line includes a second main body portion extending along the second direction, and a second protruding portion protruding from the second main body portion in the first direction and electrically coupled to a capacitance element provided corresponding to the pixel electrode via a third contact hole.
[0150] According to the configuration of Appendix 6, it is possible to achieve a configuration in which the first wiring line and the capacitance element provided corresponding to the pixel electrode are electrically coupled to each other.
Appendix 7
[0151] The electro-optical device according to Appendix 6, wherein at least the first layer and the second layer of the first wiring line are provided along an inner surface of the third contact hole.
[0152] According to the configuration of Appendix 7, the first layer provided in the third contact hole is covered with the second layer, and thus it is possible to curb etching of the first layer in the third contact hole during patterning of the first wiring line.
Appendix 8
[0153] The electro-optical device according to Appendix 7, wherein the third layer of the first wiring line flattens a recess caused by the third contact hole.
[0154] According to the configuration of Appendix 8, a film forming process is facilitated when manufacturing a layer above the third contact hole.
Appendix 9
[0155] The electro-optical device according to any one of Appendices 6 to 8, wherein the second layer and the third layer are not provided in at least a portion of the second main body portion of the first wiring line.
[0156] According to the configuration of Appendix 9, the first layer is exposed on the surface of the main body portion, and thus it is possible to increase light reflectance of the second main body portion to efficiently curb heat generation due to light absorption of the first wiring line.
Appendix 10
[0157] The electro-optical device according to any one of Appendices 6 to 9, wherein, in the second protruding portion of the first wiring line, the second layer and the third layer are not provided at a portion of the first wiring line on the main body portion side.
[0158] According to the configuration of Appendix 10, even when the areas of the second and third layers are relatively small around the third contact hole, the functions of the second and third layers are sufficiently exhibited.
Appendix 11
[0159] The electro-optical device according to any one of Appendices 6 to 9, wherein the second layer and the third layer are provided along the first direction at portions of the second protruding portion of the first wiring line and the second main body portion of the first wiring line.
[0160] According to the configuration of Appendix 11, the surplus second and third layers are reduced around the third contact hole. For this reason, it is possible to increase light reflectance by increasing the region where the first layer is exposed while improving the reliability of electrical coupling around the third contact hole.
Appendix 12
[0161] The electro-optical device according to any one of Appendices 1 to 11, wherein [0162] the first layer contains aluminum, [0163] the second layer contains titanium nitride, and [0164] the third layer contains titanium.

[0165] According to the configuration of Appendix 12, the first wiring line can be configured with a stacked structure of aluminum, titanium nitride, and titanium. For this reason, the first wiring line can reflect light by titanium covering the outermost surface, and thus it is possible to efficiently curb heat generation due to light absorption.
Appendix 13
[0166] The electro-optical device according to any one of Appendices 1 to 4, wherein a constant potential is applied to the first wiring line.
[0167] According to the configuration of Appendix 13, it is possible to curb heat generation of the first wiring line to which a constant potential is applied.
Appendix 14
[0168] An electronic apparatus including the electro-optical device according to any one of Appendices 1 to 13.
[0169] According to the electronic apparatus having the configuration of Appendix 14, the electro-optical device in which heat generation inside the substrate is curbed is provided, and thus it is possible to provide the electronic apparatus in which heat generation is curbed.

Claims

1. An electro-optical device comprising: a pixel electrode; a first wiring line including a first layer, a second layer, and a third layer; a transistor at least partially overlapping the first wiring line in plan view; and a data line provided at a layer between the first wiring line and the transistor, at least partially overlapping the first wiring line in plan view, and electrically coupled to the pixel electrode via the transistor, wherein the first layer has light reflectance and is disposed closer to the transistor than the second layer and the third layer, the second layer is provided between the first layer and the third layer and has light reflectance lower than the first layer, and the third layer has light reflectance higher than the second layer.
2. The electro-optical device according to claim 1, further comprising a relay layer including a first main body portion provided at the same layer as the first wiring line, extending along a first direction, and electrically coupled to the transistor via a first contact hole, and a first protruding portion protruding from the first main body portion in a second direction intersecting the first direction, and electrically coupled to the pixel electrode via a second contact hole, wherein the relay layer includes the first layer, the second layer, and the third layer.
3. The electro-optical device according to claim 2, wherein at least the first layer and the second layer of the relay layer are provided along an inner surface of the first contact hole.
4. The electro-optical device according to claim 3, wherein the third layer of the relay layer flattens a recess caused by the second contact hole.
5. The electro-optical device according to claim 2, further comprising a second wiring line adjacent to the first wiring line, wherein the relay layer is provided between the first wiring line and the second wiring line when seen in plan view.
6. The electro-optical device according to claim 2, wherein the first wiring line includes a second main body portion extending along the second direction, and a second protruding portion protruding from the second main body portion in the first direction and electrically coupled to a capacitance element provided corresponding to the pixel electrode via a third contact hole.
7. The electro-optical device according to claim 6, wherein at least the first layer and the second layer of the first wiring line are provided along an inner surface of the third contact hole.
8. The electro-optical device according to claim 7, wherein the third layer of the first wiring line flattens a recess caused by the third contact hole.
9. The electro-optical device according to claim 6, wherein the second layer and the third layer are not provided at at least a portion of the second main body portion of the first wiring line.
10. The electro-optical device according to claim 6, wherein, in the second protruding portion of

the first wiring line, the second layer and the third layer are not provided at a portion, on the main body portion side, of the first wiring line.

11. The electro-optical device according to claim 6, wherein the second layer and the third layer are provided along the first direction and at portions of the second protruding portion of the first wiring line and the second main body portion of the first wiring line.

12. The electro-optical device according to claim 1, wherein the first layer contains aluminum, the second layer contains titanium nitride, and the third layer contains titanium.

13. The electro-optical device according to claim 1, wherein a constant potential is applied to the first wiring line.

14. An electronic apparatus comprising the electro-optical device according to claim 1.
